PAPER_323 — CR34b Vacuum Aether Frequency Mode: F_AETHER=1.576e-35 Hz, 11th UQFF Accelerative Term

Session 93 | CompressedResonanceUQFF34bModule | 35th C++ UQFF Module FIRST occurrence of vacuum aether frequency mode in UQFF dual-channel framework

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Abstract

The CR34b module introduces the 11th UQFF accelerative term `a_aether_freq`, driven by the vacuum aether frequency constant `F_AETHER = 1.576e-35 Hz`. This term is the lowest-frequency vacuum coupling identified to date in the UQFF expansion, with a coupling coefficient of 5.253×10^{-43} — seven orders smaller than the previous minimum. It represents the cosmological-scale aether frequency mode, physically distinct from the existing resonance aether term `a_aether_res`.

Core Equation

$$a_{\text{aether\_freq}} = \frac{F_{\text{AETHER}} \cdot E_{\text{VAC,neb}} \cdot a_{\text{DPM}}}{E_{\text{VAC,ISM}} \cdot c}$$

$$= \frac{1.576 \times 10^{-35} \cdot 7.09 \times 10^{-36}}{7.09 \times 10^{-37} \cdot 3.0 \times 10^{8}} \cdot a_{\text{DPM}}$$

$$= \frac{1.576 \times 10^{-35} \times 10}{3.0 \times 10^{8}} \cdot a_{\text{DPM}} = 5.253 \times 10^{-43} \cdot a_{\text{DPM}}$$

Coupling coefficient: κ_aether_freq = 5.253×10^{-43} (smallest in UQFF expansion)

Physical Distinction: a_aether_res vs a_aether_freq

Term	Formula	Driver	Regime
`a_aether_res`	f_aether × 1e-8 × f_DPM × (1+f_TRZ) × a_DPM	f_aether (1–1000 Hz), f_DPM	System-frequency resonance
`a_aether_freq`	F_AETHER × E_VAC_neb × a_DPM / (E_VAC_ISM × c)	F_AETHER = 1.576e-35 Hz	Cosmological vacuum frequency

- `a_aether_res`: resonance mode — couples aether to DPM oscillation frequency
- `a_aether_freq`: frequency mode — couples vacuum aether to energy density ratio

Together they form the **UQFF Aether Doublet**: resonance + frequency co-sum.

System Values (a_aether_freq across CR34b 6 systems)

System	a_DPM [m/s^2]	a_aether_freq [m/s^2]
Sombrero (sys18)	~7.99e-35	~4.20e-77
Andromeda (sys19)	~6.99e-36	~3.67e-78
Universe (sys20)	~4.16e-42	~2.19e-84
Saturn (sys22)	~1.62e-24	~8.51e-67
M16 Eagle (sys23)	~1.55e-24	~8.14e-67
Crab Nebula (sys24)	~1.50e-19	~7.88e-62

Cosmic aether frequency mode is most significant for compact/high-frequency systems.

Physical Significance

- `F_AETHER = 1.576e-35 Hz` → period T = 6.35×10^{34} seconds ≈ 2.01×10^{27} years (super-Hubble oscillation)
- This is the characteristic vacuum aether oscillation at cosmological scales — beyond direct observation but present as a persistent background field in UQFF
- `LAMBDA = 1.1e-52 m`$^{-2}$ companion constant — cosmological context for aether frequency

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Classification

- **FIRST UQFF Vacuum Aether Frequency Mode** — distinct from resonance channel
- **11th UQFF accelerative term** — completes the UQFF aether doublet (res + freq)
- **Coupling coefficient 5.253×10^{-43}** — smallest identified UQFF coupling
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